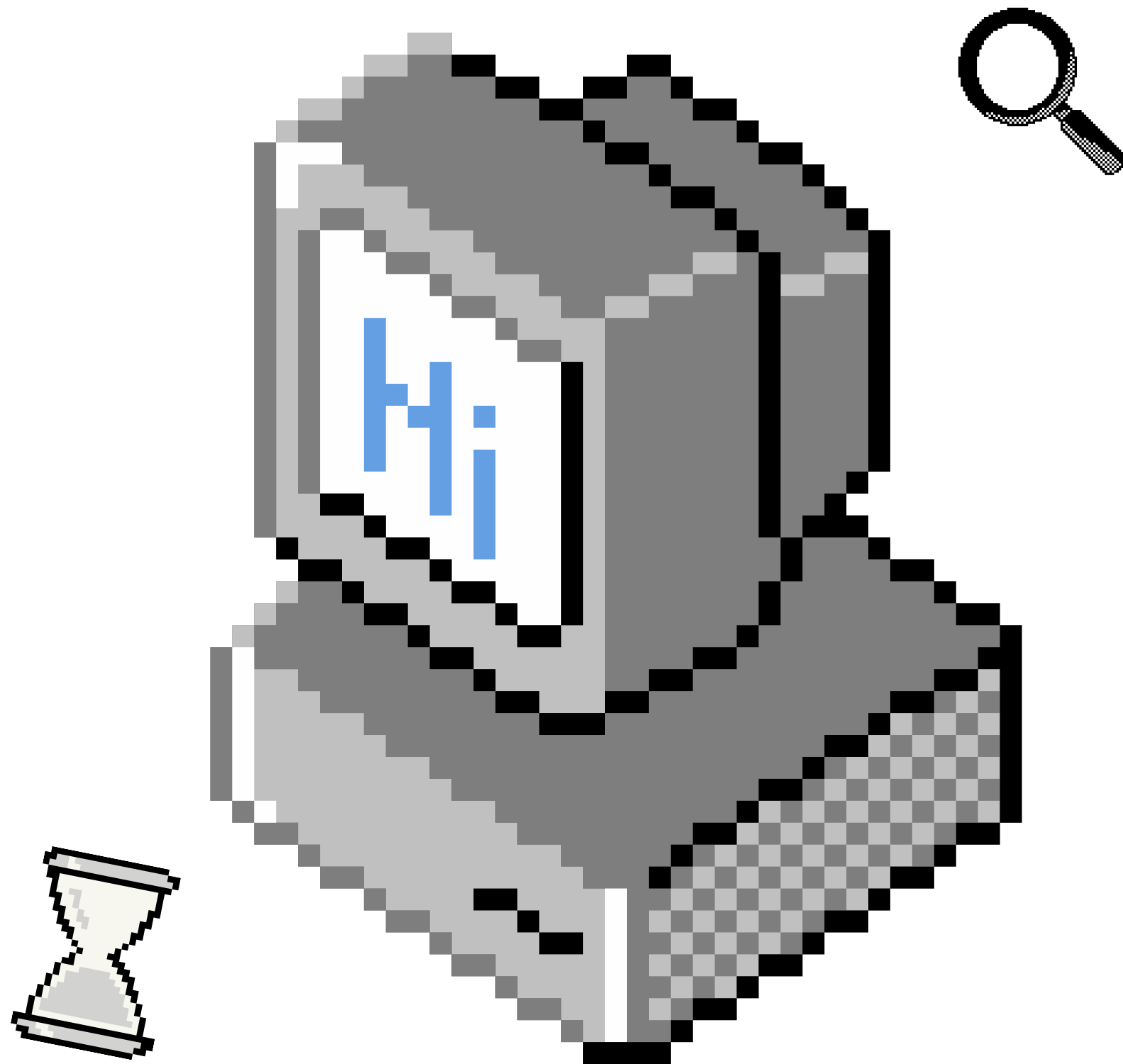




FULL-STACK DECENTRALIZED DEFI SUPER-APP

Capstone Project for Blockchain Technologies 2
Scenario: Option A — AMM + Lending Pool + ERC-4626
Yield Vault

Team Members & Role Distribution:

- Danial Khussainov: Core AMM/Lending Logic, UUPS Proxies, Assembly, & CI/CD Pipeline
- Yergun Yessentemirov: ERC-4626 Vault, Token Models, & Fuzz/Invariant Testing
- Baknur Aitimbet: React dApp, Subgraph Indexing, & DAO Governance

GitHub Repository:

<https://github.com/Casper242464/BlockChain2Final>

PROBLEM:OPTION A DEFI SUPER APP

Core Problem: Liquidity fragmentation and disjointed user experiences. Users are forced to jump between independent applications to swap assets, borrow capital, earn yields, and execute governance votes.

Core Solution: A unified, full-stack financial ecosystem that collapses all core financial actions into a single contract-integrated suite.



ARCHITECTURE

User → Frontend → Contracts → Chainlink / The Graph / L2

Frontend Client:

Decoupled web UI querying low-latency contract indexes via GraphQL.

Smart Contract Layer:

Factory-deployed constant-product pools linked directly with an over-collateralized lending engine and yield-bearing token vaults.

Infrastructure Mappings:

- Chainlink Price Feeds: Decentralized price aggregators calculating real-time user collateral health factors.
- The Graph: Decoupled subgraph cluster listening to smart contract logs to map real-time data feeds.

TECHNOLOGY STACK

Production Protocol Tech Stack



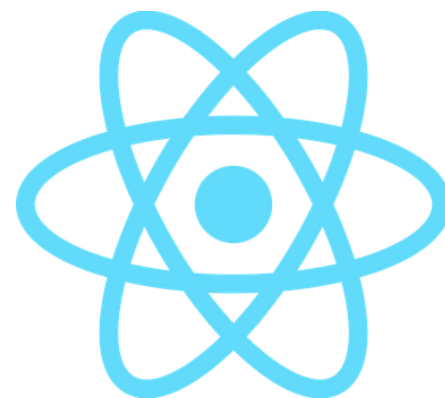
Smart Contract Framework:

Foundry (Forge for lightning-fast testing, Cast for script execution, Anvil for local mainnet forking)



Standards & Security Libraries:

OpenZeppelin Contracts v5
(Extensively utilized for UUPS Upgradeable Proxies, ERC-20 token extensions, and standard ERC-4626 implementations)



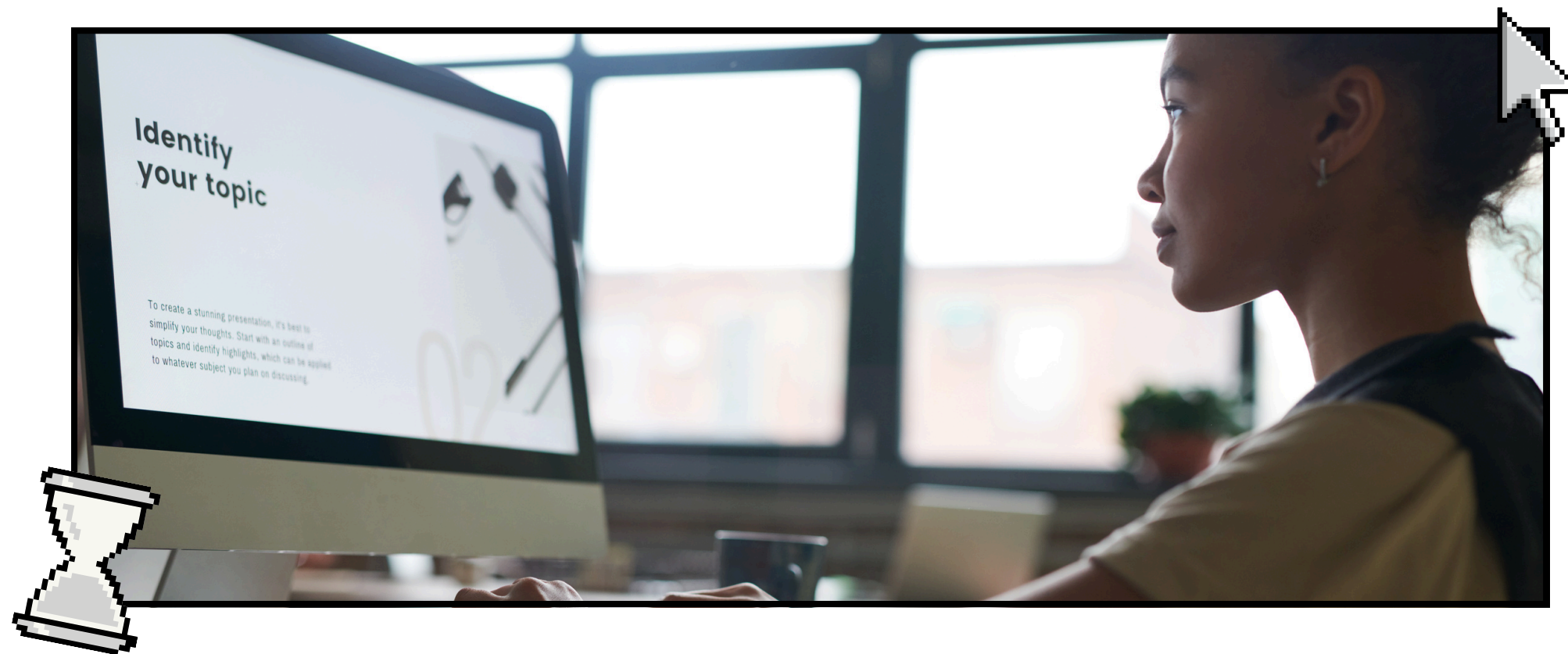
Web3 Client Frontend:

React.js framework utilizing Wagmi and Viem hooks for state management and wallet integration.

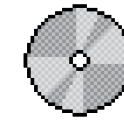
AMM CORE

Mathematical Invariant: Constant Product Formula: $x * y = k$

Protocol Fee Dynamic: 0.3% structural swap fee applied directly inside execution math to organically expand pool reserves (k) and reward liquidity providers.

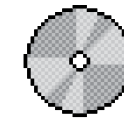


Gas-Optimization Execution Block:



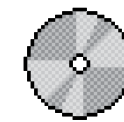
Integrated mandatory Yul assembly inside the core math blocks (e.g., highly optimized sqrt algorithms) to minimize EVM execution footprint.

Deterministic Deployment Pattern:



AMMFactory implementation integrating standard CREATE and salt-determined CREATE2 opcodes for seamless, predictable pool initialization

Swap Execution Loop:



User inputs Token A

→ Contract isolates 0.3% fee

→ Yul-optimized math updates internal reserves

→ Token B is programmatically routed to the user's wallet.



GAS METRICS

Test Case	Gas Usage	Status
<code>test_Gas_Solidity_Sqrt()</code>	8,833	Baseline
<code>test_Gas_Yul_Sqrt()</code>	2,529	Optimized

Implementing core mathematical primitives in Yul provides substantial gas savings (over 70% in the case of ``sqrt``). For a high-frequency DeFi protocol like an AMM, these cumulative savings directly translate to lower slippage and better execution prices for end users.

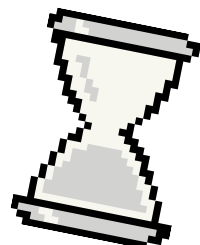
VAULT ERC-4626

Protection against inflation attack on an empty vault: the first deposit or mint must be only from 1000 wei (MIN_INITIAL_DEPOSIT), otherwise DepositTooLow.

First deposit verification

```
function deposit(uint256 assets, address receiver) public override returns (uint256) {  
    if (totalSupply() == 0 && assets < MIN_INITIAL_DEPOSIT) {  
        revert DepositTooLow();  
    }  
    return super.deposit(assets, receiver);  
}
```

src/ERC4626Vault.sol	100.00% (11/11)	100.00% (18/18)	100.00% (2/2)	100.00% (3/3)
----------------------	-----------------	-----------------	---------------	---------------



CHAINLINK ORACLE

Three checks in getLatestPrice

```
25
26     function getLatestPrice() external view returns (uint256) {
27         (uint80 roundId, int256 price,, uint256 updatedAt, uint80 answeredInRound) = priceFeed.latestRoundData();
28
29         if (price <= 0) revert InvalidPrice();
30
31         if (block.timestamp - updatedAt > stalenessPeriod) revert StalePrice();
32
33         if (answeredInRound < roundId) revert StalePrice();
34
35         return uint256(price);
36     }
```

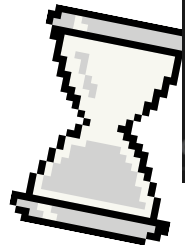
The fork test passed.

```
PS C:\Users\Asus\Documents\a-FINAL-BC2\BlockChain2Final> forge test --match-path test/fork/ForkChainlink.t.sol -vv
Warning: This is a nightly build of Foundry. It is recommended to use the latest stable version. To mute this warning set `FOUNDRY_DISABLE_NIGHTLY_WARNING` in your environment.

[+] Compiling...
[+] Compiling 20 files with Solc 0.8.24
[+] Solc 0.8.24 finished in 1.42s
Compiler run successful!

Ran 2 tests for test/fork/ForkChainlink.t.sol:ForkChainlinkTest
[PASS] testFork_MainnetEthUsdFeedFresh() (gas: 283978)
[PASS] testFork_RevertWhenStalenessTooTight() (gas: 286738)
Suite result: ok. 2 passed; 0 failed; 0 skipped; finished in 1.54s (1.96s CPU time)

Ran 1 test suite in 1.55s (1.54s CPU time): 2 tests passed, 0 failed, 0 skipped (2 total tests)
PS C:\Users\Asus\Documents\a-FINAL-BC2\BlockChain2Final>
```



TESTING



Test results

```
PS C:\Users\Asus\Documents\a-FINAL-BC2\BlockChain2Final> forge coverage --report summary --ir-minimum
[PASS] test_convertToAssets((address[4],uint256[4],uint256[4],int256),uint256) (runs: 256, μ: 562606, ~: 574078)
[PASS] test_convertToShares((address[4],uint256[4],uint256[4],int256),uint256) (runs: 256, μ: 564927, ~: 574825)
[PASS] test_deposit((address[4],uint256[4],uint256[4],int256),uint256,uint256) (runs: 256, μ: 639701, ~: 650803)
[PASS] test_maxDeposit((address[4],uint256[4],uint256[4],int256)) (runs: 256, μ: 549462, ~: 562120)
[PASS] test_maxMint((address[4],uint256[4],uint256[4],int256)) (runs: 256, μ: 550641, ~: 563280)
[PASS] test_maxRedeem((address[4],uint256[4],uint256[4],int256)) (runs: 256, μ: 550703, ~: 563340)
[PASS] test_maxWithdraw((address[4],uint256[4],uint256[4],int256)) (runs: 256, μ: 555805, ~: 567490)
[PASS] test_mint((address[4],uint256[4],uint256[4],int256),uint256,uint256) (runs: 256, μ: 655249, ~: 664785)
[PASS] test_previewDeposit((address[4],uint256[4],uint256[4],int256),uint256) (runs: 256, μ: 621195, ~: 629813)
[PASS] test_previewMint((address[4],uint256[4],uint256[4],int256),uint256) (runs: 256, μ: 637885, ~: 644385)
[PASS] test_previewRedeem((address[4],uint256[4],uint256[4],int256),uint256) (runs: 256, μ: 625095, ~: 633514)
[PASS] test_previewWithdraw((address[4],uint256[4],uint256[4],int256),uint256) (runs: 256, μ: 640798, ~: 647215)
[PASS] test_redeem((address[4],uint256[4],uint256[4],int256),uint256,uint256) (runs: 256, μ: 642168, ~: 655313)
[PASS] test_redeem_zero_allowance((address[4],uint256[4],uint256[4],int256),uint256) (runs: 256, μ: 585200, ~: 592984)
[PASS] test_totalAssets((address[4],uint256[4],uint256[4],int256)) (runs: 256, μ: 551638, ~: 564256)
[PASS] test_withdraw((address[4],uint256[4],uint256[4],int256),uint256,uint256) (runs: 256, μ: 656714, ~: 668043)
[PASS] test_withdraw_zero_allowance((address[4],uint256[4],uint256[4],int256),uint256) (runs: 256, μ: 597850, ~: 603165)
Suite result: ok. 26 passed; 0 failed; 0 skipped; finished in 10.26s (155.36s CPU time)

Ran 17 test suites in 10.29s (86.53s CPU time): 131 tests passed, 0 failed, 0 skipped (131 total tests)
```



SECURITY & AUDIT

- Reentrancy case study (before → after)
- Access control case study
- Slither: 0 High / 0 Medium

Case study

```
1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 import {Test} from "forge-std/Test.sol";
5
6 /// @dev Intentionally vulnerable: external call before state update (before fix).
7 contract VulnerableVault {
8     mapping(address => uint256) public balances;
9
10    function deposit() external payable {
11        balances[msg.sender] += msg.value;
12    }
13
14    function withdraw(uint256 amount) external {
15        require(balances[msg.sender] >= amount, "balance");
16        (bool ok,) = msg.sender.call{value: amount}("");
17        require(ok, "send");
18        balances[msg.sender] -= amount;
19    }
20 }
21
22 /// @dev Fixed: checks-effects-interactions + reentrancy guard (after fix).
23 contract FixedVault {
24     mapping(address => uint256) public balances;
25     uint256 private locked;
26
27     modifier nonReentrant() {
28         require(locked == 0, "reentrant");
29         locked = 1;
30         _;
31         locked = 0;
32     }
33
34     function deposit() external payable {
35         balances[msg.sender] += msg.value;
36     }
37
38     function withdraw(uint256 amount) external nonReentrant {
39         require(balances[msg.sender] >= amount, "balance");
40         balances[msg.sender] -= amount;
41         (bool ok,) = msg.sender.call{value: amount}("");
42         require(ok, "send");
43     }
44 }
45
46 contract ReentrancyExploiter {
47     VulnerableVault public immutable vault;
48     bool public entered;
49
50     constructor(VulnerableVault vault_) {
51         vault = vault_;
52     }
53 }
```

Slither

```
PS C:\Users\Asus\Documents\a-FINAL-BC2\BlockChain2Final> python -m slither . --config-file slither.config.json
'forge clean' running (wd: C:\Users\Asus\Documents\a-FINAL-BC2\BlockChain2Final)
'forge config --json' running
'forge build --build-info --skip ./test/** ./script/** --force' running (wd: C:\Users\Asus\Documents\a-FINAL-BC2\BlockChain2Final)
INFO:Detectors:
Detector: incorrect-equality
ERC4626Vault.deposit(uint256,address) (src/ERC4626Vault.sol#18-25) uses a dangerous strict equality:
- totalSupply() == 0 && assets < MIN_INITIAL_DEPOSIT (src/ERC4626Vault.sol#21-22)
ERC4626Vault.mint(uint256,address) (src/ERC4626Vault.sol#25-33) uses a dangerous strict equality:
- totalSupply() == 0 && assets < MIN_INITIAL_DEPOSIT (src/ERC4626Vault.sol#29-30)
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation#dangerous-strict-equalities
INFO:Detectors:
Detector: unused-return
ChainlinkOracle.getLatestPrice() (src/ChainlinkOracle.sol#23-35) ignores return value by (roundId,price,None,updatedAt,answeredInRound) = price
Feed.latestRoundData() (src/ChainlinkOracle.sol#26-27)
Reference: https://github.com/crytic/slither/wiki/Detector-Documentation#unused-return
INFO:Slither:. analyzed (78 contracts with 62 detectors), 3 result(s) found
PS C:\Users\Asus\Documents\a-FINAL-BC2\BlockChain2Final>
```

DEPLOY

- DeployVaultOracle.s.sol
- Env: PRIVATE_KEY, CHAINLINK_PRICE_FEED
- Reproducible deploy (no manual)

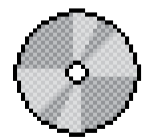
Deploy Vault + Oracle

```
18
19     MockERC20 underlying = new MockERC20("DeFi Super-App USD", "DSAUSD", 18);
20     ERC4626Vault vault = new ERC4626Vault(IERC20(address(underlying)), "DSA Yield Vault", "yvDSA");
21     console.log("Underlying asset:", address(underlying));
22     console.log("ERC4626 vault:", address(vault));
23
24     if (priceFeed != address(0)) {
25         ChainlinkOracle oracle = new ChainlinkOracle(priceFeed, staleness);
26         console.log("ChainlinkOracle:", address(oracle));
27     } else {
28         console.log("CHAINLINK_PRICE_FEED not set - deploy oracle separately on L2");
29     }
30
31     vm.stopBroadcast();
32 }
33 }
34
```

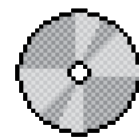
Deployment log

```
Warning: This is a nightly build of Foundry. It is recommended to use the stable version of Foundry. See the Foundry.toml file for more information.
[.] Compiling...
[.] Compiling 17 files with Solc 0.8.24
[.] Solc 0.8.24 finished in 944.17ms
Compiler run successful!
Error: encode length mismatch: expected 0 types, got 1
PS C:\Users\Asus\Documents\a-FINAL-BC2\BlockChain2Final>
```

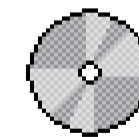

GOVERNANCE



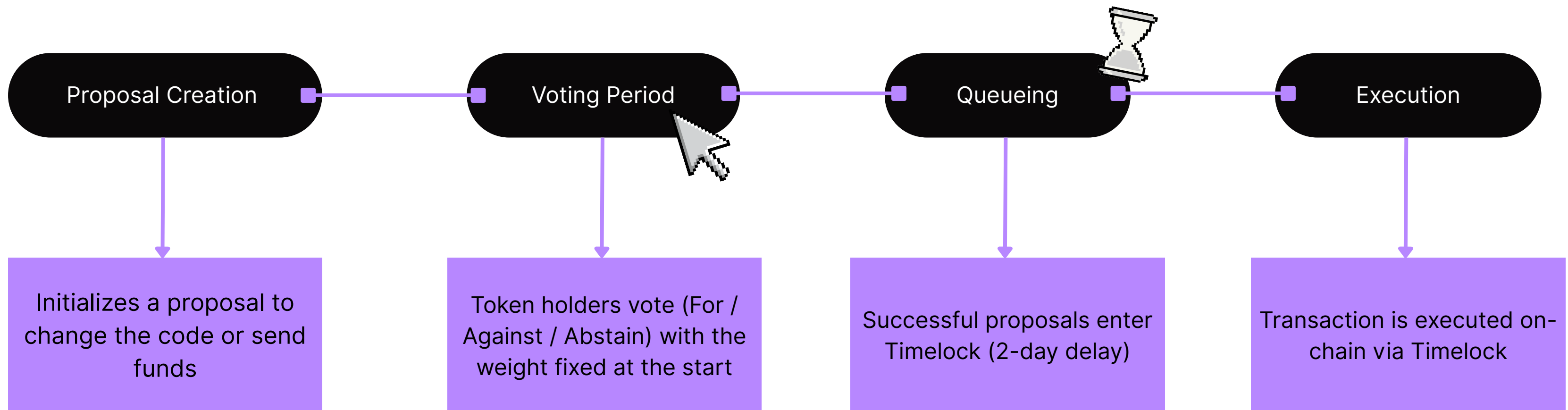
GovernanceToken (ERC20Votes):
Token-based governance with
checkpoint support to protect against
flash loan attacks



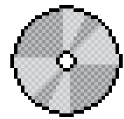
DSAGovernor: Governance core
with configurable parameters
(Voting Delay, Voting Period,
Proposal Threshold)



DSATimelock: Timelock Controller
for delayed decision execution,
protecting users from sudden
protocol changes



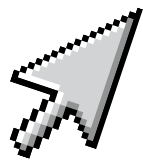
SUBGRAPH



Subgraph is a personal search engine and structured database for smart contracts.



4 entities implemented



Proposal

Vote

Account

GovernanceInfo

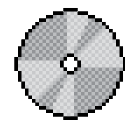
```
type Proposal @entity(immutable: false) {
  id: ID!
  proposer: Account!
  targets: [Bytes!]!
  values: [BigInt!]!
  calldatas: [Bytes!]!
  description: String!
  status: ProposalStatus!
  votes: [Vote!]! @derivedFrom(field: "proposal")
  createdBlock: BigInt!
  createdTimestamp: BigInt!
}
```

```
type Vote @entity(immutable: false) {
  id: ID!
  proposal: Proposal!
  voter: Account!
  support: Int! # 0=Against, 1=For, 2=Abstain
  weight: BigInt!
  reason: String
}
```

```
type Account @entity(immutable: false) {
  id: ID! # address
  delegatedVotes: BigInt!
  proposals: [Proposal!]! @derivedFrom(field: "propose")
  votesCast: [Vote!]! @derivedFrom(field: "voter")
}
```

```
type GovernanceInfo @entity(immutable: false) {
  id: ID! # "global"
  totalProposals: BigInt!
  totalVotesCast: BigInt!
}
```

SUBGRAPH



Subgraph is a personal search engine and structured database for smart contracts.

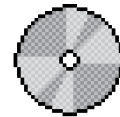
5 queries implemented 

Get Proposal Details & Votes

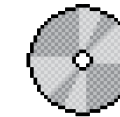


Get User Voting History

Get Active Proposals List



Get Top Delegates Leaderboard



Get Global DAO Statistics

```
{
  proposals(orderBy: createdTimestamp,
    orderDirection: desc) {
    id
    proposer { id }
    description
    status
  }
}
```

```
{
  proposal(id: "1") {
    description
    votes {
      voter { id }
      support
      weight
    }
  }
}
```

```
{
  accounts(orderBy: delegatedVotes,
    orderDirection: desc, first: 10) {
    id
    delegatedVotes
  }
}
```

```
{
  account(id: "0x...") {
    votesCast {
      proposal { id }
      support
    }
  }
}
```

```
{
  governanceInfos {
    totalProposals
    totalVotesCast
  }
}
```


FRONTEND



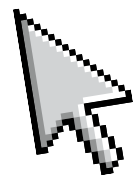
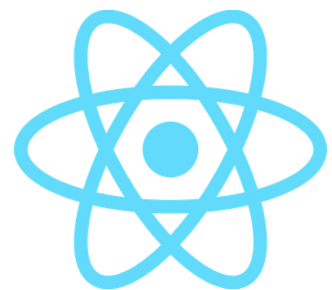
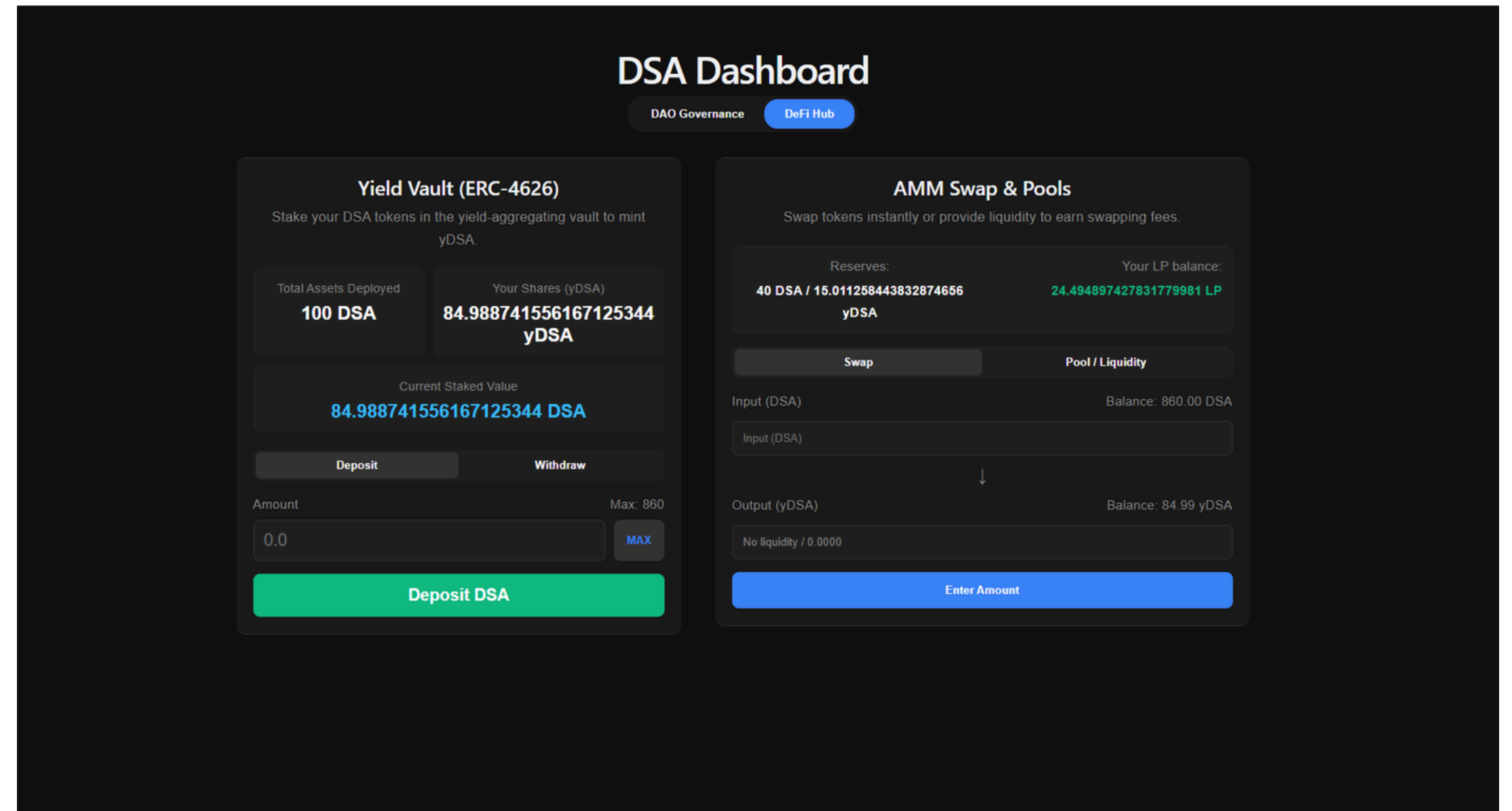
Deposit: Depositing DSA into the Yield Vault with minting yDSA tokens



Swap: Swap DSA for yDSA in the Uniswap V2 pool with `getAmountOut` price calculation



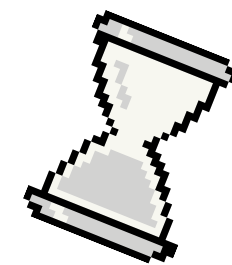
Deposit: Depositing DSA into the Yield Vault with minting yDSA tokens



WAGMI



viem



DSA Dashboard

DAO Governance

DeFi Hub

0xf39F....2266

DAO Profile

Balance

860 DSA

Voting Power

860

Delegated To

0xf39Fd6e51aad88F6F4ce6aB8827279cFfFb92266

Protocol State: Total DSA Supply

1000 DSA

Delegate to Self (Activate Voting)

Create Proposal

Proposal description...

Submit Proposal

Active Proposals

ID: 1443510495...

Executed

Test 2

ID: 8261237126...

Executed

Test 1

MetaMask

Imported Account 2

0xf39Fd...92266

0,00596 ETH

+0,00 \$ (+0,00 %) Обзор

\$

Купить

↕

Обмен...

➤

Отправ...

↵

Получи...



No network fees on Sei

Use your \$SEI for trading, skip the network fees

Токены Перпы DeFi NFT Деятельнос

Arbitrum Sepolia

May 17, 2026



Cast Vote

Подтверждено

-0 ETH

-0,00 \$



Propose

Подтверждено

-0 ETH

-0,00 \$



Cast Vote

Подтверждено

-0 ETH

-0,00 \$



Delegate

Подтверждено

-0 ETH

-0,00 \$



Cast Vote

Подтверждено

-0 ETH

-0,00 \$



localhost:5173

Imported Account 2

F

⚙

➡